

Case Study Rubric – Fake News Classification by Topic

Due:

Submission Format: GitHub Repo Link to Canvas

General Description: This case study explores how topic influences the ability to classify news articles as real or fake using data science methods.

Why am I doing this?: This assignment allows you to combine skills in data analysis, natural language processing, and machine learning to investigate a real-world problem.

What am I going to do?: You will analyze a dataset of labeled news articles, group them by topic, and evaluate how classification performance varies across topics.

How will I know I have succeeded?: You will meet expectations when your submission follows the criteria below.

<u>Category</u>	<u>Details</u>
Formatting	Submit each component listed in the rest of this rubric as advised below. ● Written Portion ○ Include in your github repo as a PDF ● Data & Code ○ Include all code in a GitHub repository ○ Repository titled: CS-FirstNameLastName ● References ○ References should be included on a separate page at the end of the Written Portion PDF file. ○ IEEE citation style
Written Portion	Discuss your interpretation of the study and thought process by following the below. ● Summarize the problem and its importance ● Describe your approach to: ○ Data preprocessing ○ Text vectorization ○ Clustering and classification ● Include a simple workflow/analysis diagram graphic ● Discuss results: ○ Compare model performance across clusters ○ Identify which topics are harder/easier to classify ● Discuss the significance of your findings ● Reflection:

	○ Discuss challenges you faced in this study ○ How you overcame them. ○ What you could have done differently ○ How you plan to improve in the future.
Code	Your code should include: ● Exploratory Data Analysis ○ Compare article and title lengths ○ Analyze vocabulary or text patterns ● Text Processing ○ TF-IDF vectorization (or similar) ● Clustering ○ K-means (or similar method) ○ Justify number of clusters ● Classification ○ Logistic regression model ○ Train on full dataset and by cluster ● Evaluation ○ Report accuracy (or similar metric) ○ Compare performance across clusters ● Include comments explaining your process ● Use the dataset in WELFake_Dataset.zip located in the DATA folder on Github (unzip the file first).
References	● Provide a list of all sources used that were not provided in IEEE format.